

- 2 Shane is studying the lengths of the tails of male red kangaroos. He takes a random sample of 14 male red kangaroos and measures the length of the tail, x m, for each kangaroo. He then calculates a 90% confidence interval for the population mean tail length, μ m, of male red kangaroos. He assumes that the tail lengths are normally distributed and finds that $1.11 \leq \mu \leq 1.14$.

Find the values of $\sum x$ and $\sum x^2$ for this sample. [6]

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(a) Test at the 10% significance level whether the data supports the inspector's suspicion. [8]

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